

TeleConnect Analytics Hub

Business & Project Overview

A Business Intelligence Case Study for the Telecommunications Industry

1. Project Introduction

TeleConnect Analytics Hub is an end-to-end Business Intelligence project developed to analyze customer behavior, revenue performance, service quality, and customer retention within the telecommunications industry. The project transforms operational telecom data into meaningful business insights using SQL, MySQL, Power BI, DAX, and Excel. Rather than presenting static reports, it delivers interactive dashboards that support data-driven decision making.

2. Business Background

Telecommunication companies generate large volumes of customer, billing, complaint, and usage data. Without a centralized analytics solution, management struggles to identify churn trends, revenue leakage, customer satisfaction issues, and high-risk customer segments. This project addresses these challenges through Business Intelligence.

3. Business Problem Statement

The organization lacked a centralized reporting solution capable of monitoring customer performance, revenue, complaints, churn, and retention. Manual reporting limited visibility and slowed business decisions.

4. Project Objectives

- Monitor customer growth and business performance.
- Measure customer churn and retention trends.
- Analyze customer demographics and segmentation.
- Evaluate revenue performance across subscription plans.
- Measure revenue loss associated with customer churn.
- Analyze complaints and service quality.
- Track customer satisfaction.
- Support strategic decision-making using interactive dashboards.

5. Analytical Approach

The project follows a complete Business Intelligence workflow including data collection, SQL analysis, data modeling, DAX measure creation, dashboard development, KPI monitoring, and business insight generation.

6. Dataset Overview

The solution uses a cleaned telecommunications dataset containing 30,000 customer records with customer demographics, subscription plans, billing, tenure, complaints, satisfaction, usage metrics, payment methods, revenue loss, churn status, and geographic information.

7. Dashboard Solution

- Home
- Executive Overview
- Customer Analysis
- Revenue Performance
- Service Quality & Complaint Analysis
- Customer Retention

8. Key Business Questions Addressed

- What is the current customer churn rate?
- Which customer segments are most likely to churn?
- Which subscription plans generate the highest revenue?
- How much revenue is lost due to customer churn?
- What complaint categories occur most frequently?
- Which customer groups should be prioritized for retention?

9. Business Value Delivered

The solution enables stakeholders to monitor KPIs, identify churn risks, improve customer retention, reduce revenue leakage, optimize subscription plans, improve service quality, and support evidence-based decisions.

10. Technologies Used

- Power BI
- MySQL
- SQL
- DAX
- Power Query
- Microsoft Excel
- GitHub

11. Skills Demonstrated

- Business Analysis
- Business Intelligence
- SQL
- MySQL
- Power BI

- DAX
- Data Visualization
- Executive Reporting
- Analytical Thinking

12. Conclusion

TeleConnect Analytics Hub demonstrates how Business Intelligence transforms operational telecom data into actionable insights that support strategic decision-making, customer retention, revenue optimization, and continuous business improvement.